



An external SSD mouse?

XPG is changing the basic gaming mouse into the XPG Vault, which is a portable solid state drive. <https://digit.in/jan22-03>



AI argues digital ethics

The Oxford Union has hosted many debaters, including an AI engine in recent times. <https://digit.in/jan22-04>

CORSAIR DOMINATOR PLATINUM RGB First Edition DDR5 RAM Kit

Blazing fast DDR5 memory is here

If you're planning on building a new Intel rig based on the latest 12th Gen Core "Alder Lake" processors, then you'll have to decide between getting a DDR4 motherboard or a DDR5 motherboard. You'll have to pick either and it goes without saying that DDR5 lends to more performance. The CORSAIR DOMINATOR PLATINUM RGB First Edition memory kit is the first DDR5 kit we've received and it upholds the build quality that Corsair is known for.

BUILD QUALITY

The heatsink design is the same as other DOMINATOR Platinum RGB memory modules. The modules are 135 mm in length, 8 mm in width and 56 mm in height. The width is not going to be a problem since most memory slots are wide enough to allow 8 mm thick modules to be placed side-by-side. On the other hand, the height is going to be a problem if you're opting for certain CPU air coolers.

You've got a two-part metal heatsink held together by screws and thermal tape. Disassembly is quite easy and we get to see a single-sided DIMM inside. CORSAIR uses their low-power CAPELLIX LEDs for the RGB lighting so the impact on the memory power sub-system isn't as much as with other brands. Aside from the LED diffuser that sits atop the DIMM, almost everything else is metal so heat dissipation isn't that big of a problem.

A closer look reveals that the RAM modules use Micron MT60B2G8HB-48B DDR5 SDRAM chips. The density for these chips is 16 Gb and they're rated for 4800 MT/s which can be overclocked to 5200 MT/s.

Corsair is using an NXP LPC82x 32-bit ARM microcontroller. The PMIC used is an APW8502C. There doesn't exist a publicly accessible data sheet for

this one so we can't delve into any details. The only thing we know for certain is that the PMIC takes 5 volts input and breaks it down into 1.1 v for Vdd and Vddq, and 1.8 v for Vpp. Separating the PMIC from the processor lends the user greater control over the overclocking capabilities of the memory chips.

Overall, the CORSAIR DOMINATOR PLATINUM RGB First Edition memory modules are built to last you a long time.

PERFORMANCE

With the DDR5 memory standard, each module can have two channels. So when you plug a pair of DDR5 modules onto your motherboard, you'll see a quad-channel configuration in your OS. We tested the units at the base profile without switching on XMP. At which point, we ran through one set of benchmarks. Without enabling XMP, the modules run at 4800 MT/s at 40-40-40-77 and with the XMP-5200 profile, this becomes 5200 MT/s at 36-38-38-74. The latency reduction might not seem like much but it's something. The pure synthetic performance bump obtained from switching to 5200



PERFORMANCE.....	90
DESIGN.....	85
VALUE FOR MONEY...	75
FEATURES.....	80

MT/s works out to be about 8.5 to 9.8 per cent.

However, when you switch to a real-world workload, the impact doesn't seem as impressive as the performance improvement is usually between 1-3 per cent.

As for power consumption, we see a noticeable increase when running

the kits at 5200 MT/s while running a heavy load. Power draw during copy operations can use close to 7.3 watts while during idle time, it goes down to about 3.2 watts.

VERDICT

This is the first DDR5 kit we've got our hands on, so there isn't much comparative data to look at. Based on what we've seen so far, the build quality of the CORSAIR DOMINATOR PLATINUM RGB DDR5 modules is like no other. We'll update the review on our website once we get our hands on additional RAM kits.

—Mithun Mohandas

SPECIFICATIONS

SPD LATENCY: 40-40-40-77 | SPD SPEED: 4800MHz | SPD VOLTAGE: 1.1V | PERFORMANCE PROFILE: XMP 3.0 | TESTED LATENCY: 36-36-36-76 | TESTED VOLTAGE: 1.25 | TESTED SPEED: 5600.

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